

Lab 3 - Exercises for Pipes and Redirection

Here are some exercises that need to be done, after unzipping the files uploaded on Moodle. These files are extracted from the `/var/log` directory on veridian. System administrators frequently need to examine these files, and do some analysis of the health of the system. Unzip the files into a directory called "lab-3" and work inside it.

Exercise 1: Filtering output

Use the `ls` command to list all files in the current directory, and then pipe the output to `grep` to find only files that have "login" in their name

```
ls | grep "login"
```

Exercise 2: Counting lines

As expected, the "failed-logins" file keeps a record of login attempts that failed. Use the `cat` command to display the contents of a file called "failed-logins", and then pipe the output to `wc` to count the number of lines in the file.

Exercise 3: Sorting output

Use the `ls` command to list all files in the current directory, and then pipe the output to `sort` to sort the files in alphabetical order.

Exercise 4: Finding unique lines

Use the `cat` command to display the contents of the file called "successlogin.log", and then pipe the output to `sort` and then to `uniq` to find only the unique lines in the file.

Exercise 5: Extracting information

Use the `ps` command to display information about running processes. Now read the man page for `ps` and find the option to list all running processes on the system

Exercise 6: Converting output

Use the `date` command to display the current date and time, and then pipe the output to `tr` to convert the output to uppercase.

Exercise 7: Formatting output

Read the man page for the command called `column`. Use the `ls` command to list all files in the current directory, and then pipe the output to `column` to format the output into columns.

Exercise 9: Removing duplicates

Use the `cat` command to display the contents of the file called "failed-logins", and then pipe the output to `sort` and then to `uniq -u` to remove duplicate lines and only show unique lines. Count the number of lines in the file, when you *do* use `uniq` and when. you *do not* use `uniq`.

Exercise 10: Counting occurrences

VIM is a much-loved editor that is used on unix-like systems. Use the `cat` command to display the contents of the file called "alternatives.log". Search for the phrase "Vim" and count the number of lines that it occurs on. Modify your search to make the search process case-insensitive (*hint*: check out the man page for `grep` to find out how to make the search case insensitive) and now count the number of lines it occurs on.

Here are some exercises that require the use of input and output redirection:

Exercise 1: Redirecting output to a file

Use the `ls` command to list all files in the current directory, and then redirect the output to a file named `file_list.txt`.

Exercise 2: Appending output to a file

Use the `date` command to display the current date and time, and then append the output to a file named `log.txt`.

Exercise 3: Redirecting input from a file

From the `/tmp` directory on 'veridian', copy the file called `numbers.txt` to your `lab-3` directory. Use the `sort` command to sort the contents of this file, and then redirect the output to a new file named `sorted_numbers.txt`.

Exercise 4: Using input and output redirection together

Use the `cat` command to display the contents of a file named `alternatives.log`, and then pipe the output to `grep` to find only lines that contain the word `vim`, and then redirect the output to a file named `output.txt`.

Exercise 5: Redirecting error messages

Use the `ls` command to list all files in a non-existent directory, and then redirect the error message to a file named `error.log`.

Exercise 6: Using the `tee` command

Read the man page for the command called `tee`. Use the `ls` command to list all files in the current directory, and then pipe the output to `tee` to display the output on the screen and also redirect it to a file named `file_list.txt`.

Exercise 7: Using the `>>` operator to append output

Use the `date` command to display the current date and time, and then append the output to a file named `log.txt` using the `>>` operator.

Exercise 8: Using the `2>` operator to redirect error messages

Use the `ls` command to list all files in a non-existent directory, and then redirect the error message to a file named `error.log` using the `2>` operator.

Exercise 9: Using the `&&` operator to redirect both output and error messages

Use the `ls` command to list all files in a non-existent directory, and then redirect both the output and error messages to a file named `output_and_error.log` using the `&&` operator.

These exercises should help you practice using input and output redirection in bash to manipulate and process output from various commands.
